

Manual

Controlling of Articles/Scrap BDE-CAA 8.2

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1 Overview Controlling article/scrap

Purpose

The overview controlling article / scrap function package offers the ability to evaluate data entered as a part of shop floor data collection (BDE). You use the function package if:

- You would like to evaluate the shop floor data entered relating to articles
- You would like to evaluate the scrap data entered.

Integration

The evaluations are based on the data collected during production. For this purpose, the article-based evaluations aggregate the quantities and times relating to an order, while the scrap evaluations analyze the scrap quantities recorded.

Features

- Article profile
 - Article profile with comparison to the scrap accumulated as well as the time recorded (consolidated in resource performance accounts)
- Scrap statistics
 - Workplace and article-related scrap (reason) statistics. Tabular display of total and scrap quantity, scrap rate and the calculated scrap ratio. Graphic display of scrap quantities by workplace, article or scrap reason.
 - Scrap statistics expanded with columns like machine, machine designation, cost center, group, article, order, tool, scrap reason, scrap quantity, scrap rate as % of total quantity (formula), total quantity, scrap quantity as % of total scrap quantity
 - Ability to select by machine, order, article number, cost center, group, tool, time and shift
 - Optional calculation of scrap ratios as % of total quantity, of articles or machines
 - Display of yield, rework, problem quantity and total quantity
- Scrap profile
 - Evaluations of scrap quantities entered over time. Staging of data in table and pivot table form. Data evaluation in pivot table form by various criteria. Graphical presentation of data in pivot table form.

2 Article Profile

Summary

Menu	Order Management → Order Controlling → Article Profile
Transaction code	artpf
Function authorization	artpf

The Article profile application is designed for users in the departments Production controlling, Order controlling or Final costing. It compares orders across a defined period of time that produced the same article and displays comparisons of their production times and downtimes. Based on the selection made via the order header article (the order's final product) and, optionally, a defined period of time, all orders that produced that article are displayed along with their results.

Integration

The application is based on the order-related quantities and times entered during production.

Requirement

The application only looks at production orders - not individual operations. Only those production orders are considered that have the control status "finished" in their order headers. Production orders with a different control status are not considered.

Selection criteria

The application provides the selection criteria listed below: Irrespective of these selection criteria, generally only orders are selected that have already started (in process) or that are finished.

Final product

This selection criterion refers to the article in the order header. All orders are displayed, to which the article entered has been assigned. The use of wildcards (placeholders *) is allowed.

Article designation

This selection criterion refers to the article designation defined in the order. All orders are displayed that contain the article designation selected.

Order status

This selection criterion refers to the status of the order. Only those orders are displayed with an order status that matches the criteria entered. Irrespective of this restriction, generally only orders are considered and selected that have already started (in process) or that are finished.

Category

This selection criterion refers to the order type category of the order. Only those orders are displayed with an order type that is associated with the category entered.

Order

This selection criterion refers to the order number. Orders are displayed that contain the order number entered. There is an option to use wildcards (placeholders *).

Order type

This selection criterion refers to the order type. Orders of the order type entered are displayed.

Order end ... until ...

This selection criterion refers to the point in time at which the order(header) is changed to "finished" status. In this case, only the finished orders are displayed with an actual end that is set between the selected dates. This occurs irrespective of whether there is a restriction on the order status.



The responsibility area is not checked in this application.

Article profile detail application (table)

All orders that match the selections entered are displayed in the article profile detail application table view. A selection of columns is described below:

Status

Current status of the order. Generally, only orders are considered that have started or that are finished.

Status since date, status since time

Point in time as of which the current status of the order applies.

Order

Order number of the order

Final product

Article number of the order(header).

Article designation

Name of the article listed in the order(header).

SUT, DCI, SCI, LCI, IMS, IMN, SET, STA, U8, U9, MUT

The durations that were posted to each resource performance account are displayed in the columns.

Retention period of order

The retention period of the order is the period between the time the order was first transferred from the PPS system ("order release" = order header creation date in HYDRA) and the time of the actual logoff (in terms of time) of the last active operation of the order.

Please note: Whether the time of the order transfer is the time it was first transferred from the PPS system or the time it was re-transferred because it had been deleted in the meantime for technical reasons, cannot be determined or taken into consideration using HYDRA. If an order is transferred more than once, and with each transfer the previous order is deleted, the creation date of the order header is the time of the last transfer.

Please note: If the order is not yet finished, the value calculated here is not significant.

Lead time

The order duration is the period between the time the first operation of the order is logged on and the time the last active operation is logged off.

Please note: If the order is not yet finished, the value calculated here is not significant.

Processing time

The processing time of the order is the sum of the main utilization times (RPA 11) of all recordable active operations.

Downtime period

The downtime period of the order is the sum of the downtimes (RPA 1..6, RPA 8..10) of all recordable active operations.

Please note: the setup time can be found in the corresponding resource performance account (RPA 7/ SET).

Assignment time

The occupancy time is the sum of the setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1..6, RPA 8..10) of all recordable active operations.

Yield

Recorded yield in base quantity unit of the last recordable operation. This requires that the unit in which the quantity is entered (primary quantity unit) can be converted into the base quantity unit.

Scrap

Sum of the scrap quantities entered for all operations posted in base quantity unit.

Rework

Sum of the rework quantities entered for all operations posted in base quantity unit.

Open quantity

Sum of the recorded problem quantities entered for all operations posted in base quantity unit.

Unit

Quantity unit of the order

Rate of capacity utilization

The rate of capacity utilization is the ratio of processing time (RPA 11) to occupancy time (RPA 1..11)

Setup rate

The setup rate is the ratio of setup time (RPA 7) to occupancy time (RPA 1..11)

Article profile detail application (graphic)

By selecting one or more orders in the table, the following information is displayed in the graphic for those orders:

Upper graphic

In the upper graphic of the article profile, three bars are displayed for each of the orders selected in the table:

- On the left: Scrap in percent based on the finished total quantity of the order (yield + scrap + rework + problem quantity in base quantity unit).
- In the middle: Main utilization time in percent based on the occupancy time of the order
- On the right: Downtime periods (RPA 1 to 10) in percent based on the occupancy time (RPA 1 to 11) of the order

The colors of the time bars correspond to the defined RPA colors; scrap bars are generally displayed in red.

Lower graphic

In the lower graphic of the article profile, the absolute values of a single RPA or of the scrap are displayed in bar form for each selected order. The user determines which value is displayed by clicking in the upper graphic.

The individual order numbers are displayed on the X axis, while the unit of the value is displayed on the Y axis. If durations are greater than 24 hours, they are displayed in days.hours:minutes:seconds.

Each value selected (scrap or name of the resource performance account) is displayed beneath the graphic.

3 Scrap Statistics

Overview

Menu	Order management → Order controlling → Scrap statistics
Transaction code	scrst
Function authorization	scrst

Any person in a company who wants to reduce scrap must have knowledge of how and where scrap is accrued. The *Scrap statistics* provide an overview of all scrap reasons and scrap quantities that were recorded in a specified period of time. The report also includes the scrap that has been recorded automatically. The *Scrap statistics* use graphic displays to show ranking lists of the most frequent scrap reasons.

Integration

The order-related log records of record type "T" are the data basis used to display quantities in the scrap statistics. The log records are generated on the shop floor terminal when part quantities are manually uploaded or when an operation is interrupted or logged off and quantities are recorded.

Selection criteria

The application provides the following selection criteria:

Workplace/short name/group/cost center/company/responsibility area

These fields specify the workplaces/machines that are used for the evaluation. To select an entry, you access the master data of the workplace/machine configuration.

Article/article designation

Selection by article or article designation

Order/order type

You can limit the evaluation to one order or order type.

Date

Enter a period of time to narrow down the log records displayed. The application selects the log records of record type "T" with a posting time that is included in the period of time specified.

Shift

In the period of time specified above, only those log records are selected that are assigned to the shift specified.

Time

In the period of time specified above, only those log records are selected that were booked in the time specified.

Reason

Selection based on a specific scrap reason.

Sales order

If a sales order is assigned to the production order, this sales order can be used to select the data.

Project number

If a project number is assigned to the production order, this number can be used to select the data.

Report group

Use the option *Report group* to narrow down the selection to a specific group of workplaces/machines. You must define the report group in the [group configuration](#).

Show yield

If this option is enabled, the yield recorded in the specified period of time is also displayed for the selected operation. Additional entries are added to the table. Note: the column *Yield* is not displayed by default. Use the column configurator to show the column.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the following function authorizations: scrstPerf + chkresp.

Detail application *Scrap statistics*

The detail application *Scrap statistics* provides an overview of all scrap reasons and scrap quantities that were recorded in a specified period of time. The data selected is based on the order-related log records of record type "T". The quantities identified are always displayed in the primary quantity unit. To total the quantities, the system does not convert any quantity units.



The *Scrap statistics* only show operations that had at least one upload of a part quantity including scrap in the specified evaluation period.

Note: the column *Yield (P)* only displays yield quantities, if the option *Show yield* is enabled. Yield is shown in a separate row for each order/OP and workplace.

Notes on specific columns:

Scrap

The column *Scrap* shows the scrap quantity with reference to the scrap reason, the workplace and the order/operation.

Note: the scrap quantity shown can also be a negative quantity in certain cases. For example, this can be the case if scrap is offset against yield.

Scrap rate

Share of scrap (primary quantity unit) in the total quantity produced (primary quantity unit) of the operation in the evaluation period.

The total quantity is equal to the total of all four quantity accounts (yield, scrap, rework, open quantity). Note: rework quantity and open quantity are not shown in the *Scrap statistics*.

Calculation:

$$\text{Scrap rate} = 100.0 * \text{EGR_AUS} / (\text{ANR_GUTP} + \text{ANR_AUSP} + \text{ANR_NCHP} + \text{ANR_PRBP})$$

Detail application *Comparison: yield - scrap*

This detail application provides a comparison of the yield and scrap quantity (both in primary quantity unit). The comparison is shown in a bar chart. The bars show the ratio of the two quantity types. The percentage is visualized for each bar.



Note: the yield displayed is calculated using the total of all yield quantities of the operations identified. If the selection includes several process steps (workplaces), the yield quantities of all workplaces selected are totaled. It does not matter if several operations of one order are then included.

Use the application [Order-related statistic](#) to get an overview of one order only.

Detail application *Scrap ranking list (scrap reasons)*

The *Scrap ranking list (scrap reasons)* shows the scrap quantities recorded for the different scrap reasons in a graphic form.

The entries selected in the tabular detail application control the display of the bar chart. To show all entries of the table in the graphic, select the complete table. To select the complete table, click the table field in the top left corner. To select entries, we recommend to disable the option *Show yield*.

Each bar of the bar chart shows the totaled value for each scrap reason. The total is calculated using all entries selected. The bar also shows the percentage of this scrap reason in the total scrap quantity recorded in the evaluation period. This total scrap quantity is displayed in the total line of the *Scrap statistics* detail application.

Example: the detail application *Scrap statistics* shows a total scrap quantity of 1,171 in the total line for a selected period. If you select a single entry with a scrap quantity of 122, the graphic shows a percentage of 14.42.

Detail application *Scrap ranking list (workplaces)*

The *Scrap ranking list (workplaces)* shows the scrap quantities recorded for the different workplaces/machines in a graphic form.

The entries selected in the tabular detail application control the display of the bar chart. To show all entries of the table in the graphic, select the complete table. To select the complete table, click the table field in the top left corner. To select entries, we recommend to disable the option *Show yield*.

Each bar of the bar chart shows the totaled value for each workplace. The total is calculated using all entries selected. The bar also shows the percentage of this scrap reason in the total scrap quantity recorded in the evaluation period. This total scrap quantity is displayed in the total line of the *Scrap statistics* detail application.

Toolbar



Order information

Calls the [Order information](#) for the currently selected order.



Failure mode analysis (function authorization faep)

Click this button to call the [Failure mode analysis](#).

4 Scrap profile

Overview

Menu	Order management → Order controlling → Scrap profile
Transaction code	scrf
Function authorization	scrf

Any person in a company who wants to reduce scrap must have knowledge of how and where scrap is accrued. The *Scrap profile* provides an overview of all scrap reasons and scrap quantities that were recorded in a specified period of time.

Selection criteria

The application provides the following selection criteria:

Date

Enter a period of time to narrow down the log records displayed. The application selects the BDE log records of record type T with a posting time that is included in the period of time specified.

Shift

In the period of time specified above, only those log records are selected that are assigned to the shift specified.

Time

In the period of time specified above, only those log records are selected that were booked in the time specified.

Workplace/group/cost center/company/short name/designation/responsibility area

These entries specify the workplaces/machines used for the evaluation. To select an entry, you access the master data of the workplace/machine configuration.

Report group

Use the option *Report group* to narrow down the selection to a specific group of workplaces/machines. You must define the report group in the [group configuration](#).

Reason

Selection based on a specific scrap reason.

Order/order type

You can limit the evaluation to one order.

Article

Selection according to the article

Sales order

If a sales order is assigned to the production order, this sales order can be used to select the data.

Project number

If a project number is assigned to the production order, this number can be used to select the data.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.



If the selection period exceeds the period of time of the online data area, the system implicitly selects the data of the medium-term data area. You need not explicitly activate the access to the medium-term data area.

Condition: the data retention periods of orders (object ANR) and of log records (object ADEPRO) must be identical.

Detail application *Scrap profile*

The detailed report *Scrap profile* provides a table including all scrap reasons and scrap quantities that were recorded in a specified period of time.

Note on the column *Scrap rate*:

The scrap rate refers to the total quantity booked in the selected period of time (total of yield, scrap, rework quantity and open quantity in primary quantity unit) for the combination workplace/MES order number.

Example: in a specified period of time, the following quantities have been recorded at a workplace for an order/operation:

Day	Yield	Scrap
1	10	1
2	12	0
3	0	0
4	14	1
5	8	2

If you select the period of time Day 2 - Day 5, then the days 4 and 5 show scrap quantities that are displayed in the *Scrap profile*. The scrap rate refers to the total quantity booked in the selected period of time (here: $12+14+8+1+2 = 37$):

Day	Scrap	Scrap rate
4	1	2.7
5	2	5.4



To calculate the scrap rate, the system also uses postings that only include a yield, but no scrap. Entries in the result line with a scrap quantity of 0 are therefore possible.

You can hide these entries using the *Filter editor*.

Detail application *PivotGrid*

The detailed report *PivotGrid* provides an overview of all scrap reasons and scrap quantities that were recorded in a specified period of time. According to the settings of the pivot table, the application shows e.g. one column for each scrap reason. The scrap recorded automatically is displayed separately.

Scrap (P)

This pivot element includes the primary quantities.

Scrap reason

This pivot element includes the scrap reasons (number of the scrap reason).

Date

This pivot element includes the dates when the scrap was recorded.

The detail application contains not only the fields mentioned here, but numerous other fields. You can select these fields using the field list. Right-click the area above the column headers to open the context menu and select *Show field list*.



To calculate the scrap rate, the system also uses postings that only include a yield, but no scrap. Entries in the result line with a scrap quantity of 0 are therefore possible.

You can hide these entries using the *Filter editor*.